

SPLUNK AGENTIC OPS HACKATHON · PLATFORM & DEVELOPER EXPERIENCE

DualContext

Production incident. Two contexts. One answer.
45 seconds, not 13 minutes.

Splunk MCP

SigMap MCP

Hosted Models

Groundedness-scored

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THE PROBLEM

Incidents need **two** kinds of context

When production breaks, a developer needs operational reality **and** code structure — and today they live in two different tools.

Splunk

knows the symptom

47 errors, line 47, 8× baseline — but not *which file or method* owns it.

IDE

knows the code

The class and method — but not *which error* is firing in production right now.

13 min

tab-switching

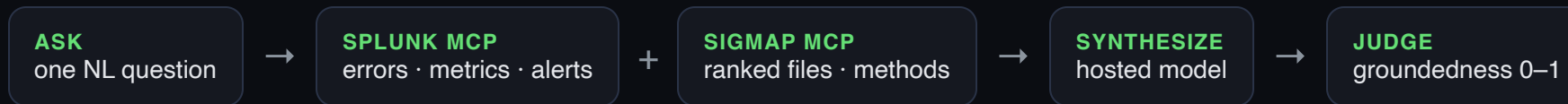
Average investigation, alt-tabbing between dashboard and editor. Mental overhead.

Neither source alone is sufficient. **Together they pinpoint both.**

THE SOLUTION

One query, **two MCP servers**

DualContext fans one developer question out to two MCP servers in parallel, then a Splunk hosted model fuses the answer.



Single-source tools stop here

“Here are your logs.” Or: “here’s your code.” The developer still has to bridge the two by hand.

DualContext bridges them

Cites a specific log *and* a specific file+method in one grounded answer — scored for groundedness every time.

HOW WE USE SPLUNK

The MCP Server is the protagonist

CAPABILITY	HOW DUALCONTEXT USES IT	PRIZE
MCP Server	Primary integration — operational context via run_splunk_query (handshake, Bearer auth)	Best Use of MCP
Hosted Models	gpt-oss synthesizes both contexts into one grounded answer	Best Use of Models
SPL	Errors, error-rate timechart, alert history — three searches per query	Core
SigMap MCP	Second MCP server — ranked code structure at ~97% token reduction	Composability

MCP as an **interop layer** — exactly what Splunk is promoting. Splunk does the heavy lifting; SigMap adds the code lens.

PROOF — RUNNABLE, SCORED

Grounded answer in ~1 second

Two MCP servers in parallel, fused and judged — query: *“Auth service throwing 47 errors per hour. What’s causing it?”*

```
SPLUNK MCP · 3 SPL searches, 47 events (1061 ms)
SIGMAP MCP · ranked 3 files, 180 tokens vs 45000 raw (99.6% reduction)
PARALLEL · both ready in 1061 ms (not 1862 ms serial)
SYNTHESIS · gpt-oss-120b fused both sources
JUDGE · groundedness 0.83 PASS (threshold 0.7)
```

Answer cites `JwtTokenProvider.validateToken()` (line 47) *and* the Splunk error spike — both, in one paragraph.

THE GROUNDEDNESS GUARANTEE

Every answer is scored

A groundedness judge scores how well the answer cites the actual files/methods — no silent hallucination.

WITH BOTH CONTEXTS

0.83

Cites the specific log **and** the specific file+method. PASS.

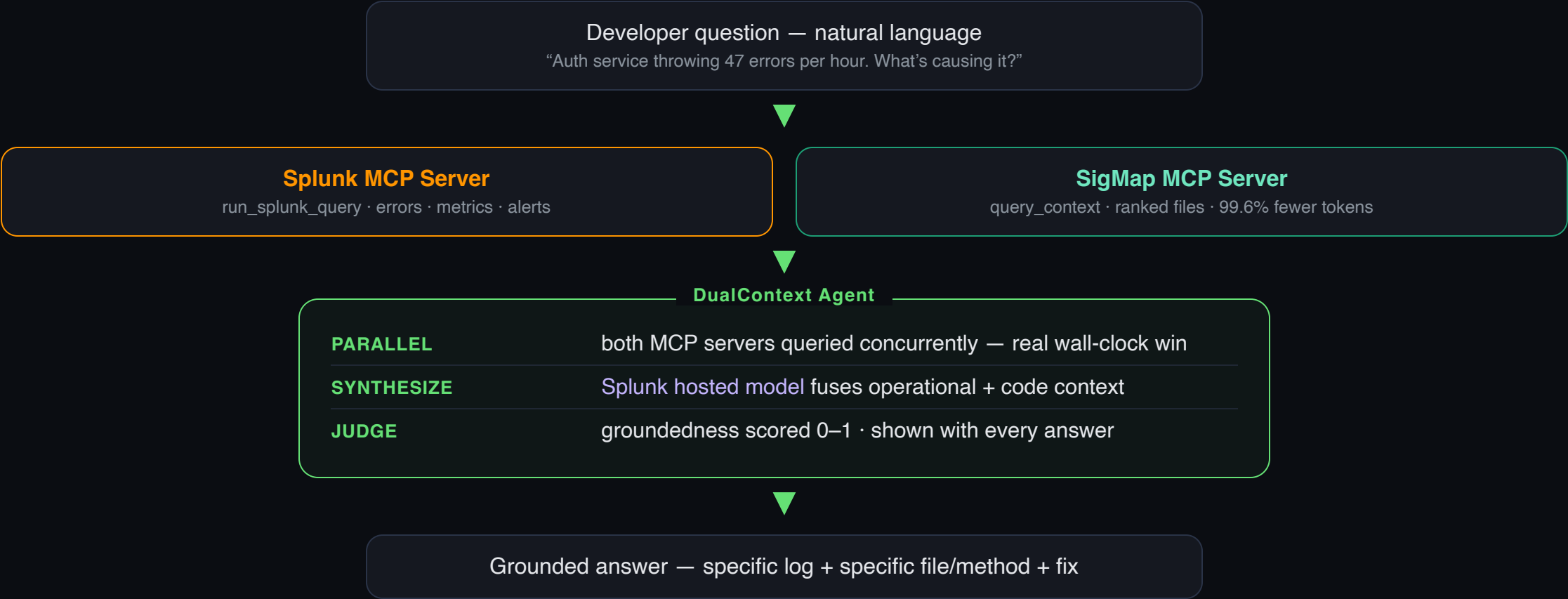
GENERIC ANSWER

0.15

“Spring Security typically uses...” — no codebase grounding. FAIL.

The developer sees the score with **every** response. **Trust, made visible.**

One query · dual MCP · fused answer



WHY IT WINS

Judged on four criteria

Technological Implementation

Two real MCP servers composed, hosted-model synthesis, parallel execution, groundedness scoring. Runs in ~1s.

Quality of the Idea

Dual-MCP fusion — the most architecturally sophisticated way to use Splunk's MCP Server as an interop layer.

Potential Impact

Every developer investigates incidents. 13 min → 45 s, with a groundedness guarantee on every answer.

Design

Dual-source investigation UI — Splunk green + SigMap teal, evidence panels, groundedness badge.

Squarely on the **Platform & DevEx** mandate: *simplify Splunk interactions.*

WHY NEITHER SOURCE ALONE IS ENOUGH

Splunk + code, or you're guessing

Splunk alone

"NullPointerException at line 47" — *which file?*
You still open the IDE.

SigMap alone

"JwtTokenProvider.validateToken()" — *which error, when?* You still open Splunk.

Together

Specific file + specific error + root cause + fix.
One grounded answer, scored.

In the demo incident, **neither source alone was sufficient** — Splunk had the line, SigMap had the method; only together did they pinpoint the bug.

TWO CONTEXTS. ONE ANSWER.

DualContext

Splunk MCP + SigMap MCP, fused by a Splunk hosted model. Grounded, every time.

Splunk MCP

SigMap MCP

Hosted Models

Groundedness

SPL

github.com/manojmallick/dualcontext